

RingWatch — Ring -3 Intelligence Report

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System: Intel Core Ultra 5 125H (Meteor Lake-P) | HP

ME Firmware: CSME v18.1.15.2544 (current) / v18.0.10.2351 (backup)

ME OS: MINIX 3 microkernel | 30+ modules

ME CPU: Intel Quark x86 32-bit (estimated 100-400 MHz, not documented by Intel)

ME Memory: Isolated SRAM inside SoC (estimated 1-5 MB)

SPI Flash: 32 MB total, ME region 8.4 MB (AES-256-CBC encrypted)

1. ME Firmware State

Field	Value
Device State	ENABLED
Firmware Version (current)	0:18.1.15.2544
Firmware Version (backup)	0:18.0.10.2351
Operation State	NORMAL
Firmware Init Complete	YES
ME Powered	NO
Recovery Mode	YES
Runtime Mode	NO
Error Code	2321
Feature Subset	1280
HBM Version	2.2
TX Queue Limit	50

2. HECI Clients (MEI Interface)

Total active HECI clients: 9

#	Name	UUID	Description	Action	Driver
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					Max Conn	Max Len	
1	ME_CLIENT	6861ec7b-d07a-4673-856c-7f22b4d55769	Unknown ME client	RESEARCH	0	4096	none
2	ME_CLIENT	dd17041c-09ea-4b17-a271-5b989867ec65	Unknown ME client	RESEARCH	1	4096	none
3	ME_CLIENT	082ee5a7-7c25-470a-9643-0c06f0466ea1	Unknown ME client	RESEARCH	1	8196	none
4	ME_CLIENT	42b3ce2f-bd9f-485a-96ae-26406230b1ff	Unknown ME client	RESEARCH	0	12288	none
5	ME_CLIENT	309dcde8-ccb1-4062-8f78-600115a34327	Unknown ME client	RESEARCH	1	4096	none
6	MKHIF_FIX	55213584-9a29-4916-badf-0fb7ed682aeb	OS version telemetry — POISON TARGET	POISON	0	4096	none
7	GSC_PROXY	0f73db04-97ab-4125-b893-e904ad0d5464	Graphics Security Command Proxy	SAFE	1	8192	mei_gsc_p
8	PTT	8e6a6715-9abc-4043-88ef-9e39c6f63e0f	Platform Trust Technology / fTPM — DO NOT TOUCH	PROTECT	1	4096	none
9	AMT	8c2f4425-77d6-4755-aca3-891fdbc66a58	Active Management Technology — POISON TARGET	POISON	1	12288	none

Poison Targets (ME clients feeding data to ME — to be blocked/spoofed)

Name	Description	UUID
MKHIF_FIX	OS Version Telemetry	55213584-9a29-4916-badf-0fb7ed682aeb
AMT	Active Management	8c2f4425-77d6-4755-aca3-891fdbc66a58

Protected Clients (DO NOT TOUCH — security critical)

Name	Description	UUID
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PTT	Platform Trust (fTPM)	8e6a6715-9abc-4043-88ef-9e39c6f63e0f
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3. PCI Device — CSME HECI #1

Field	Value
PCI Address	00:16.0
Device	Intel Meteor Lake-P CSME HECI #1 (rev 20)
Vendor:Device ID	0x8086:0x7E70
Subsystem	HP Device 8CAC (0x103C:0x8CAC)
IOMMU Group	11
Kernel Driver	mei_me
Memory Region	0x402028b000 (64-bit, 4KB)
Interrupt	Pin A routed to IRQ 160
Power State	D0 (active), PME(D3hot+)
MEI Driver Size	61440 bytes (60 KB)
MEI Core Size	184320 bytes (180 KB)

4. HECI MMIO Registers

MMIO Base: 0x402028b000 (4KB)

Access: BLOCKED by CONFIG_STRICT_DEVMEM — registers would show live buffer fill levels

Offset	Register	What It Shows
0x00	H_CB_WW	Host Circular Buffer Write Window
0x04	H_CSR	Host Control Status — ready, interrupt, buffer pointers
0x08	ME_CB_RW	ME Circular Buffer Read Window
0x0C	ME_CSR_HA	ME Control Status — ready, interrupt, buffer pointers

5. SPI Flash

Field	Value
Device	/dev/mtd0ro (read-only)

Total Size	32 MB
ME Region	0x00004000 – 0x0086A000 (8.4 MB)
Encryption	AES-256-CBC (keys derived from hardware fuses)
Access	Read-only via OS, ME has read-write access

6. Power Consumption (RAPL)

Domain	Energy (kJ)	Utilization	Notes
Package (total)	236910.0	90.37%	Total CPU package
Core	134592.2	51.34%	CPU cores only
Uncore (includes ME)	2278.2	0.87%	Best indirect ME activity signal
ME Estimate	~100-200 mW	—	Drawn from 3.3VSB rail, not separately measurable

7. MEI Driver Memory

Driver	Size	Ref Count	Notes
mei_me	60 KB	1	Main HECI driver
mei_core	180 KB	3	Core MEI subsystem (depends on mei_me, mei_gsc_proxy)
mei_gsc_proxy	16 KB	0	Graphics Security Command Proxy

8. Network Security — AMT Port Scan

Port	Service	Status
623	ASF/RMCP	CLOSED
664	ASF	CLOSED
16990-16996	AMT Web Interface	CLOSED

AMT is NOT active — no remote management ports listening.

9. USB WiFi Adapter (ME-Free Network Path)

Field	Value
Interface	wlx00c0ca989edb
Driver	ath9k_htc

Chip	Atheros AR9271 (USB 2.0)
Firmware	v1.4
Bus	USB 3-2:1.0
Signal	-42 dBm
TX Bitrate	72.2 MBit/s MCS 7 short GI
RX Bitrate	72.2 MBit/s MCS 7 short GI
RX Bytes	117,890,759
TX Bytes	192,601,941
RX Packets	228,293
TX Packets	204,316
RX Dropped	0
TX Dropped	86

10. Internal WiFi (NC-SI Endpoint)

Field	Value
Interface	wlo1
Chip	Realtek RTL8852BE (PCIe 01:00.0)
Carrier	DOWN (no carrier)
Driver	UNBOUND (rtw89_8852bte removed)
Radio (phy0)	KILLED (rfkill soft-block)
NC-SI Status	ISOLATED — no network, no radio, no driver, no credentials
NetworkManager	Unmanaged
udev Rule	99-disable-wlo1.rules (forces down at boot)

11. Defense Architecture Summary

Layer	Protection	Status
IOMMU	intel_iommu=on iommu=pt intel_iommu=strict	PERMANENT (GRUB)
Internal WiFi	Driver unbound + rfkill + udev + NM unmanaged	PERMANENT
USB WiFi	ath9k_htc driver pinned, authorization locked	ACTIVE

MAC Sync	PIF identity rotation → USB WiFi MAC change	ACTIVE (every 30min-3h)
RAM Zero	Memory flush every 30min + boot wipe	ACTIVE
xHCI Monitor	PCI config space tamper detection	ACTIVE (every 5min)
HECI Monitor	Client activity monitoring	ACTIVE
Port Monitor	AMT ports 623/664/16990-16996	ACTIVE (7-day capture)
RingWatch	Real-time Ring -3 dashboard	ACTIVE (port 8849)
GRUB Safety	5-second menu timeout	PERMANENT

12. Ring Hierarchy

Ring	Component	Our Access	ME Access
Ring 3 (Userspace)	Applications, browser, PITBULL	Full	Via HECI/DMA
Ring 0 (Kernel)	Linux kernel, drivers, eBPF	Full	Via HECI/DMA
Ring -1 (Hypervisor)	None (no VM)	N/A	N/A
Ring -2 (SMM)	System Management Mode	NONE	Full (firmware-level)
Ring -3 (ME/CSME)	Intel Quark + MINIX 3	NONE (HECI only)	FULL (hardware-level)

Ring -3 operates below the OS with its own CPU, OS, memory, and firmware. We cannot directly access or control it — only monitor via HECI and limit its network/DMA access.